

Dre. Athénaïs Gautier - Research Engineer



Contact

✉ athenais.gautier@onera.fr
💻 <https://athenaisgautier.github.io/>

Languages

🇫🇷 French - Native
🇬🇧 English - Fluent

Research interests

Spatial statistics Uncertainty quantification
Gaussian processes Inverse problems
Reproducing Kernel Hilbert Spaces

Programming

R Jupyter Notebooks Markdown Python

PROFESSIONAL EXPERIENCE

Feb. 2025 -
Present

Adjunct Lecturer

📍 France

Teaching Assistant at Master's level, primarily in English:

- ◇ At ENSAE (Palaiseau): tutorials for Theoretical Foundations of Machine Learning and Simulation and Monte Carlo Methods.
- ◇ At ENSAI (Rennes): supervision of a methodological project in statistics and AI.

Oct. 2024 -
Present

Research Engineer

📍 Palaiseau, France

ONERA

Research in uncertainty quantification and multidisciplinary optimization

- ◇ Surrogate modelling, active learning, and optimization under uncertainty for vehicle design.
- ◇ Design of optimized experiments and databases dedicated to surrogate-model construction.
- ◇ Sensitivity analysis and uncertainty quantification for computationally expensive numerical simulations.
- ◇ Co-development and successful funding of a PhD project (2027) on robust drone navigation using RKHS representations (with K. Dahia).
- ◇ Supervision of two research internships (Gaussian processes, RKHS embeddings) and coordination of an ONERA working group on kernel/RKHS methods.

Sept. 2023 -
June 2024

Postdoctoral Researcher

📍 Montréal, QC, Canada

McGill University

Research in statistics and geostatistics:

- ◇ Probabilistic modelling of mineral resources.
- ◇ Integration of expert knowledge and real-world drilling data.
- ◇ Development of C++ code.

Nov. 2018 -
June 2023

Research and Teaching Assistant

📍 Bern, Switzerland

Institute of Mathematical Statistics and Actuarial Science, University of Bern

Until May 2020, the affiliation was shared with Idiap Research Institute (Martigny, Switzerland).

Research Assistant in Statistics

- ◇ Nonparametric estimation of conditional probability distributions and uncertainty modelling for complex systems.
- ◇ Stochastic inverse problems in interdisciplinary collaborations in the natural sciences.
- ◇ Adaptive sampling for optimization and inversion.
- ◇ R programming and development of a CRAN package.

Senior Consultant at the Institute of Mathematical Statistics and Actuarial Science (2021 - 2023)

- ◇ Statistical analyses for industry and academic partners.
- ◇ Communication of results and knowledge transfer to non-statistical audiences.
- ◇ Work with financial, social, and biomedical data.

Teaching Assistant at Master's level, primarily in English:

- ◇ At the Institute of Mathematical Statistics and Actuarial Science, University of Bern: optimization, linear models, spatial statistics, R programming, and statistics for climate sciences.
- ◇ At Idiap Research Institute and UniDistance: Foundations of Statistics for AI (2019 and 2020).
- ◇ Co-supervisor (main supervisor: Prof. Dr. D. Ginsbourger) of the Master's thesis "Gaussian process regression on molecules: some performance assessments and comparisons". (2021).

Mar. 2018 - Oct. 2018	Research Intern <i>Idiap Research Institute</i> Interdisciplinary collaboration with the Centre de Recherches Énergétiques et Municipales. ♦ Mixed-integer optimization of the energy mix of a holiday village. ♦ Gaussian-process surrogate modelling. ♦ Sensitivity analysis of the system.	📍 Martigny, Switzerland
June 2017 - Aug. 2017	Engineering Intern <i>ELM Leblanc</i> ♦ Study of data integration in energy systems. ♦ Contribution to the development of an industry-sponsored PhD project.	📍 Drancy, France

🎓 EDUCATION

Nov. 2018 - May 2023	PhD in Statistics <i>University of Bern</i> Until May 2020, the affiliation was shared with Idiap Research Institute (Martigny, Switzerland). Thesis: “Modelling and predicting distribution-valued fields with applications to inversion under uncertainty”, supervised by Prof. David Ginsbourger. Graduated with the highest possible grade and the Latin distinction <i>summa cum laude</i>.	📍 Bern, Switzerland
Sept. 2017 - Oct. 2018	MSc in Applied Mathematics <i>Université Paris Dauphine</i> Specialization in <i>Statistical and Financial Engineering</i> , statistics track. Graduated with highest honours and ranked first in the cohort.	📍 Paris, France
Sept. 2015 - Sept. 2018	Engineering Degree <i>École des Mines de Saint-Étienne</i> Major in Statistics and Data Science. Ranked in the top 10% of students in these specializations.	📍 Saint-Étienne, France
Sept. 2013 - July 2015	Preparatory Classes MPSI/MP <i>Lycée Henri IV</i>	📍 Paris, France

📖 PUBLICATIONS

Journal	SLGP: An R Package for Spatial Conditional Density Estimation, A.G. , Submitted in July 2026	
Proceedings	Spatial Logistic Gaussian Processes based surrogates for spatially dependent discrete outputs: modelling, uncertainty quantification, and data acquisition., A.G. , Accepted - MODA 14 proceedings, Statistical Papers, 2026.	
Journal	Targeted CRPS-based sequential design with application in chemical space, L. Friedli, A.G. , A. Broccard, D. Ginsbourger, Accepted - Journal of Computational and Graphical Statistics (JCGS), 2026.	
R Package	SLGP: implementing spatially dependent density estimation in R, A.G. , On CRAN	
Journal	Continuous logistic Gaussian random measure fields for spatial distributional modelling, A.G. , D. Ginsbourger, Accepted in Annals of the Institute of Statistical Mathematics	
PhD Thesis	Modelling and predicting distribution-valued fields with applications to inversion under uncertainty, A.G. , Accepted by the Faculty of Science in May 2023	
Proceedings	Goal-oriented adaptive sampling under random field modelling of response probability distributions, A.G. , D. Ginsbourger and G. Pirot, Published in ESAIM: Proceedings and Surveys.	
Workshop	Probabilistic ABC with Spatial Logistic Gaussian Process, A.G. , D. Ginsbourger and G. Pirot, Published in Third Workshop on Machine Learning and the Physical Sciences, NeurIPS 2020.	

★ AWARDS AND DISTINCTIONS

-
- **Best PhD Student Oral Presentation Award:** GdR MASCOT-NUM Annual Meeting 2022
 - **Workshop Organization Grant:** Fund for the Promotion of Young Researchers, University of Bern 2022
 - **Student Travel Award:** SIAM Conference on Uncertainty Quantification 2022
 - **Best PhD Student Poster Presentation Award:** GdR MASCOT-NUM Annual Meeting 2021
 - **Louis Neltner Excellence Scholarship** from the Fondation de l'École des Mines de Saint-Étienne, awarded on the basis of academic excellence. 2016
 - **Odon Vallet Excellence Scholarship**, awarded on the basis of academic excellence. 2014
 - **French National Merit Scholarship**, awarded following the highest distinction in the French Baccalauréat. 2013-2015

ACADEMIC SERVICE

Member of the Steering Committee of the **RT-UQ** *from Fall 2026*

Core member of the organizing committees for the following events:

- **Lifting Inference with Kernel Embeddings 2023** (webmaster) *June 26-30, 2023*
- **Current Frontiers in Gaussian Processes** (lead organizer) *Aug. 24-26, 2022*
- **University of Bern-IMT Toulouse Workshop** (lead organizer) *May 3-4, 2022*
- **Lifting Inference with Kernel Embeddings 2022** (webmaster) *Jan. 10-14, 2022*

Peer reviewer for the following venues and journals:

- **Artificial Intelligence and Statistics (AISTATS),
Conference on Neural Information Processing Systems (NeurIPS),
Sensitivity Analysis of Model Output (SAMO).**
- **Journals: Technometrics, Journal of Multivariate Analysis.**
- **Workshops: Machine Learning and the Physical Sciences at NeurIPS,
Synergy of Scientific and Machine Learning Modeling at ICML.**